SM-2123: Engineering Physics I Syllabus
Dr. German Colón - Fall 2026

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Office (Student) Hours: MF 1:00-1:50p, Th 10:00-10:50a; or by appointment
Course Website: Accessed via Blackboard

Prerequisites: SM-1212 – Calculus I; **Corequisites**: SM-2113 – Calculus II

Course Description:
An introduction to the fundamental principles of physics with an emphasis in rigid-body mechanics. This course focusses on the classical theory of motion as understood using Newton's laws and the conservation principles of energy and momentum. These laws and principles describe the motion of both everyday objects and celestial bodies.

Course Materials:
1. [Recommended] **Textbook:** Moebs, W., Ling, S. J., & Sanny, J. (2016). *University Physics Volume 1*. OpenStax.
   Access for free at: https://openstax.org/details/books/university-physics-volume-1
2. [Recommended, not required] **Textbook:** Knight, R. D. (2022). *Physics for Scientists and Engineers, Volume 1* (5th ed.). Pearson.
3. [**Required**] **iClicker Response System**. You may use iclicker+, iclicker2, or iclicker Gen1. You may purchase or rent it.

Learning Outcomes:
At the end of this course each student should be able to:
1. Perform appropriate dimensional and unit analysis.
2. Perform vector analysis in one and two dimensions.
3. Apply the concepts of kinematics to various problems where acceleration is constant.
4. Apply Newton's Law of motion to analyze and solve various problems involving acceleration, gravitation, and friction in one and two dimensions.
5. Analyze the relationship between work, energy, and power.
6. Use knowledge of momentum and impulse to analyze and solve problems in one and two dimensions.
7. Use knowledge of uniform circular motion to analyze and solve various rotational problems.
8. Use knowledge of force, torque, and equilibrium to solve various problems.
9. Demonstrate a sufficient understanding of the required mathematics (including differential and integral calculus) that allows solutions to various problems to be obtained.

10. Demonstrate an understanding of the applicable physics by assessing the accuracy and correctness of all results.
11. Verify the theoretical ideas and concepts covered in lecture by completing a host of experiments.
12. Apply the scientific method to experiments performed in the laboratory.
13. Develop procedural and observational skills as experimental data is taken.
14. Apply analytical techniques and graphical analysis to data taken during an experiment.

Grading:

The expected breakdown of your grade will be as follows:

Laboratory	15%	
Pre-Lectures	2.5%	(bonus)
Supplemental Activities	14%	
Partial Exams	51%	
Final Exam	20%	
Total	102.5%	

Students often ask for their "average" grade during the semester but this term is misleading because different components have different weights. Instead, students should consider their *weighted average*, named *earned score* in this class, which is the number of points earned up to that moment out of their goal of 100 points at the end of the semester.

Final letter grades will be determined using the following guideline:

A:	93.10 – 100.0%	**C**:	73.10 – 76.99%
A-:	90.00 – 93.09%	**C-**:	70.00 – 73.09%
B+:	87.00 – 89.99%	**D+**:	68.51 – 69.99%
B:	83.10 – 86.99%	**D**:	66.51 – 68.50%
B-:	80.00 – 83.09%	**D-**:	65.00 – 66.50%
C+:	77.00 – 79.99%	**F**:	< 65.00%

The final grade, <u>before</u> any of the penalties described below, is calculated as:

FinalGrade = 0.15*LabGrade + 0.025*Pre-Lectures + 0.14*SupplementalActivities + 0.51*PartialExams + 0.20*FinalExam.

Course Components and Expectations (Summary)

The student evaluation comprises four partial exams, one final exam, the laboratory, and supplemental activities which include quizzes and class participation. Class attendance is mandatory, absences require documentation, and there are strict requirements to excuse absence from an exam. There is no tolerance for academic dishonesty. No cell phones permitted during class.

Course Components and Expectations (Details)
Blackboard will be our main hub to access many components of the course. There will be content areas dedicated to the posting of: announcements and reminders, lecture slides, lesson videos, handouts and worksheets, and assessments.

Lesson Delivery and Attendance: Lessons will be delivered in-person. Attendance is mandatory and will be taken at the beginning of each class. It is your responsibility to be on time and participate in every class. Each *unexcused* absence will result in a decrease in your final grade by 0.5%.

The following observations are further made:

- Absences must be supported by appropriate <u>documentation</u>. Examples include medical notes, court orders, obituaries, and academic approvals by the Dean. Statements like "family obligations, family emergency, personal situation, etc." on their own, are not considered excused absences without documentation.
- To excuse absence from an exam, the documentation must provide enough information to determine that: (1) you were sick, injured, or otherwise unable to attend <u>at the time</u> of the exam; and (2) it would have been detrimental to your performance, or it would not have been possible, to take the exam under those conditions. Without this, you are not eligible for a make-up exam.
- A Special Liberty is considered an agreement between *the Regiment* and *the student*, and not extending to academics. A "special lib" on its own, even if signed by me, is <u>not</u> deemed an excused absence.
- Every three late arrivals to class will be treated as an unexcused absence.
- Frequently stepping out of the classroom for long periods of time, or sleeping in class, will be treated like a late arrival.

Lectures: During lectures we will discuss the main ideas of the material listed in the schedule. To fully understand the material, it is necessary to read each specified section and/or to watch the posted videos.

Classroom Etiquette: Students are expected to be respectful of others and to practice common courtesy. These behaviors enhance your ability to learn in this class, and to avoid distractions and disruptions to others.

Use of Electronic Devices:

1. The use of cell phones, smart phones, smart watches, or other mobile communication devices is disruptive and disrespectful. Therefore, these devices must be <u>**turned off**</u> and <u>**stowed away**</u>. Violations will affect your *Participation and Engagement* grade (described on page 4), and may result in disciplinary actions.
2. The use of laptops in class is not permitted; while the use of tablets for note-taking is strongly discouraged. Neither is suitable for this course. If you have a compelling reason for using your laptop or tablet during class, and I approve its use, please sit in one of the designated areas of the lecture room (e.g. first row).

Laboratory: ALL students are required to participate in the laboratory, even if they are repeating the course.

Note: Failure in the lab component of the course automatically triggers FAILURE of the course!

Pre-Lectures: To assess readiness for class, most lessons include pre-lecture assignments based on the reading or the Panopto video(s) assigned for the day (see the schedule). Pre-Lectures will be accessed via Blackboard. They will be due right before class starts; late pre-lectures will not be accepted. Blackboard questions will be graded on correctness; embedded video questions will be graded on completion. This is a bonus component where you may earn up to 2.5% on your final grade.

Supplemental Activities: Several other types of activities will contribute to your grade including iclicker questions, quizzes, worksheets, and supplemental problems. All students are expected to participate and complete these exercises.

- *iclicker Questions (6/14)*: Frequently (almost daily) I will ask iclicker questions in class. These items will be graded on completion or correctness, depending on the question. These activities cannot be made up. Make sure to bring your iclicker to every lesson.
- *Class Participation and Engagement (2/14)*: This component assesses your class participation and engagement beyond iclicker submissions. Factors that have a positive impact include asking questions, offering responses and participating in discussions; while those that affect negatively include the use of unapproved electronic devices (see Classroom Etiquette), sleeping in class, frequently stepping out of the classroom, and other distractions.
- *Quizzes (4/14) and Supplemental Problems (2/14)*: These activities may be administered as either in-class or take-home assessments. They will be graded on completion or correctness, depending upon the activity. Some in-class activities cannot be made up. Please email me a picture related to forces by the second Friday of the semester, to show that you have read the syllabus. This will be worth five points on your quizzes grade.

Review the Policy on Academic Integrity regarding non-tolerated actions.

Homework: There will be daily homework assignments. They will be posted on Blackboard at least one week prior to their due date. While this component earns you no credit, it is your opportunity to test your knowledge, and assess your understanding and exam preparedness. This problem-solving practice is necessary to perform well in exams (see remarks under Other Notes and Expectations).

Students are strongly discouraged from using online platforms/tools like Chegg, ChatGPT, or Google to assist them in the completion of the assignments.

Homework Notebooks: Students are encouraged to work out homework problems in a notebook or loose-leaf binder. This is good practice on presenting proper physics solutions (see Policy on Problem Solving).

Partial Exams: There will be four fifty-minute cumulative partial exams administered during class time.

The exams will include material covered in lectures, assigned readings, labs and supplemental problems. The partial exams comprise 51% of your final grade. Exams will be weighted by score: your highest scoring exam will be worth a greater percentage (e.g. 11%, 13%, 13%, 14%). Exam grades are **not** scaled.

You will be allowed to bring a note card to exams and to use a calculator. The note card should be no larger than 8″ x 5″ and it may only contain formulas and definitions; worked-out examples, or results derived from them, are **<u>not</u> allowed**. You will submit your note card along with your exam.

Check the exam schedule so that you do not miss any of the exams. In general, there will be **<u>no</u>** make-up exams. Exceptions to this rule will be considered on a case-by-case basis, and immediate communication is <u>required</u>. While appropriate documentation justifying your absence is required for a make-up exam, it does not guarantee it (see Lesson Delivery and Attendance section).

Review the Policy on Academic Integrity regarding non-tolerated actions. Violations will result in stricter examination policies and possibly, in failure of the course.

Final Exam: There will be one final cumulative exam during finals week. The scheduling of final exams is done by the Academy, not by your instructor. Students are required to be available for their exam during the stated time. Please note that vacations, previously purchased tickets, misreading the schedule, etc., are not viable excuses for missing the final exam.

A grade of at least 40% in the final is necessary to **pass the class**. The final exam assesses retention of the material.

Review the Policy on Academic Integrity regarding non-tolerated actions.

Deficiencies: Determination of a deficiency in the course will be based on the grades of the partial exams completed at the time.

Policy on Problem Solving and Grading: An important distinction is made between the terms "solution" of and "answer" to a problem. The **solution** refers to the set of steps that describe the way, process or strategy utilized to achieve an outcome. The **answer** is generally the last step in the solution; it shows a final expression or numerical value.

For many homework questions, only the answer is requested or given. However, in all submitted work (e.g. exams, quizzes, worksheets) it is the **<u>solution</u>** what is graded. The (final) answer is only worth about 10-15% of the problem.

Students often argue that they are penalized for not following the instructor's method. This view is generally incorrect, based on confusing answer with solution. To earn full credit in a problem, the **solution** must be complete, clear, and with correct physics grammar. It is possible to get a 'correct' answer by chance or with flawed physics/ mathematics.

Simply stated, a correct (final) answer with missing or unclear steps earn little to no credit; while an incorrect answer with proper solution may earn most credit.

Policy on Academic Integrity:

Adherence to the Honor Code is expected. Cheating, plagiarism or any other type of academic dishonesty will not be tolerated. In all assessments, especially exams and quizzes, you may not share your problems, solutions, thoughts, ideas or any comment with another student or post them on any online/offline platform. You may not use electronic devices (e.g. cell phones, AI glasses), textbooks, answer keys or any online sources/tools such as Google, Chegg, ChatGPT or similar sites, to search for the solutions. These actions are considered violations of Academic Integrity.

Any evidence of academic misconduct will result in a score of zero for that assessment. Depending on the severity of the infraction, it may result in failure of the course and the incident to be reported to the Dean. I may also refuse your return to the classroom.

Submitted work where I suspect academic misconduct (e.g. AI assistance) will not be graded until evaluated by external reviewers, such as the Department Chair and the Dean, and a recommendation from them is received.

Academic dishonesty will prevent you from mastering the course content. Being forthright in your academic endeavors will prepare you for life as a professional in your field.

Policy on Diversity and Inclusion:

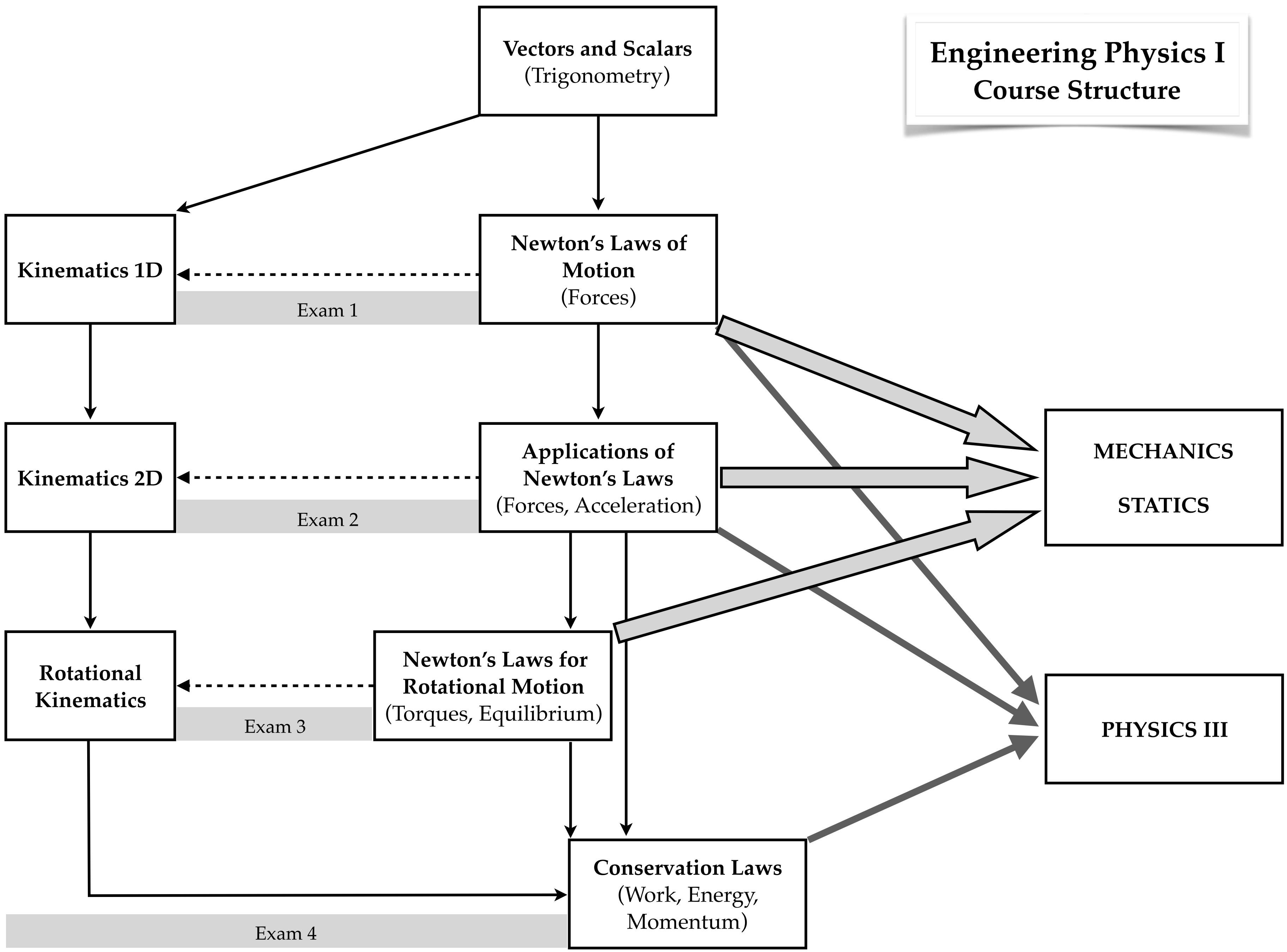
It is my intent to present materials and activities that are respectful of diversity: gender, sexuality, disability, age, socioeconomic status, ethnicity, race, and culture. Your suggestions are encouraged and appreciated. Please let me know ways to improve the effectiveness of the course for you personally or for other students or student groups.

Disability Compliance:

MMA is committed to providing academic accommodations to students who qualify. Students who had an IEP or 504 Plan in high school, or others who believe they may need and qualify for accommodations in this class are encouraged to contact the Academic Accessibility Services Coordinator, Dr. Elaine Craghead (ADAcompliance@maritime.edu or x5350), ideally within the first week of class. Please remember that academic accommodations are not retroactive.

Other Notes and Expectations:

- Physics is a subject that requires students to be active learners. As a result, all students must take notes during class and actively participate in problem solving.
- Some students believe that for an introductory class of this nature, their previous knowledge of physics and mathematical skills will be sufficient to get them through the class without much work. While this may be true, it is uncommon. If you are not putting a concerted effort into reading the textbook, completing the homework, and understanding the material, your grades will demonstrate that lack of effort.
- A skeletal outline of my lecture slides (with information intentionally left out) for the semester are posted on Blackboard/Course Website. They should complement but not substitute taking notes in class. Consider printing them and completing the missing sections in class.
- Students often complain that they do not see enough example calculations in class. While it is important to see examples, you will learn more from actually working through example problems yourself than by watching someone solve problems.
- Tutoring is available by faculty and students at the Academic Resource Center (ARC); make sure you take advantage of this.
- While this is a general outline of this course, I reserve the right to make any changes to the syllabus as I see fit. An up-to-date version of the syllabus will be posted on Blackboard/Course Website.
- The graphic syllabus on the next page illustrates the course structure, how the topics are interconnected and for which courses they will be relevant. Make sure to address points of confusion early on by attending or scheduling office hours.



Engineering Physics I
Course Structure
Vectors and Scalars
(Trigonometry)
Kinematics 1D
Newton's Laws of Motion
(Forces)
Exam 1
Kinematics 2D
Applications of Newton's Laws
(Forces, Acceleration)
Exam 2
Rotational Kinematics
Newton's Laws for Rotational Motion
(Torques, Equilibrium)
Exam 3
Conservation Laws
(Work, Energy, Momentum)
Exam 4
MECHANICS
STATICS
PHYSICS III

Engineering Physics I - Fall 2026 - Dr. Colón

Lec	Day	Date	Material Covered	Reading Assignments	Pre-Lecture Due	Homework Due
1	W	2-Sep	Course Introduction	Syllabus		
2	F	4-Sep	Trig Review; Vectors: Addition	Appendix E, Ch 2.1	PL #02 Due before class	
	M	7-Sep	**No Class - Labor Day**			
3	W	9-Sep	Vectors: Components	Ch 2.2	PL #03 Due before class	HW #03 Due at 11:59 pm
4	F	11-Sep	Introduction to Forces	Ch 2.3, 5.1	PL #04 Due before class	HW #04 Due at 11:59 pm
5	M	14-Sep	Newton's 1st Law: Inertia, Free-Body Diagrams	Ch 5.2, 5.6-5.7	PL #05 Due before class	HW #05 Due at 11:59 pm
6	W	16-Sep	Newton's 1st Law: (Translational) Equilibrium	Ch 5.2, 5.7, 6.1	PL #06 Due before class	HW #06 Due at 11:59 pm
7	F	18-Sep	Motion, Kinematic Variables	Ch 3.1-3.3	PL #07 Due before class	HW #07 Due at 11:59 pm
8	M	21-Sep	Instantaneous Values, Constant Acceleration	Ch 3.3-3.4	PL #08 Due before class	HW #08 Due at 11:59 pm
9	W	23-Sep	Units and Dimensional Analysis	Ch 1.2-1.4	PL #09 Due before class	HW #09 Due at 11:59 pm
10	F	25-Sep	Constant Acceleration: Free-Fall	Ch 3.5	PL #10 Due before class	HW #10 Due at 11:59 pm
11	M	28-Sep	Newton's 2nd Law	Ch 5.3, 6.1	PL #11 Due before class	HW #11 Due at 11:59 pm
12	W	30-Sep	Dynamics: Mass, Weight, Weightlessness	Ch 5.4, 6.1	PL #12 Due before class	HW #12 Due at 11:59 pm
E 1	F	2-Oct	***Exam #1: Lessons 1-11***			
13	M	5-Oct	Newton's 3rd Law: Interactions, Action/Reaction	Ch 5.5	PL #13 Due before class	HW #13 Due at 11:59 pm
14	W	7-Oct	Newton's Laws: Defining Your System	Ch 6.1, 5.7	PL #14 Due before class	HW #14 Due at 11:59 pm
15	F	9-Oct	Dynamics: Friction	Ch 6.2	PL #15 Due before class	HW #15 Due at 11:59 pm
	M	12-Oct	**No Class - Columbus Day**			
16	T	13-Oct	Motion in 2D: Projectiles (Monday Schedule)	Ch 4.1-4.3	PL #16 Due before class	HW #16 Due at 11:59 pm
17	W	14-Oct	Motion in 2D: Projectiles	Ch 4.3	PL #17 Due before class	HW #17 Due at 11:59 pm
18	F	16-Oct	Motion in 2D: Circular Motion (Relative Mot. Optional)	Ch 4.4 (4.5)	PL #18 Due before class	HW #18 Due at 11:59 pm
19	M	19-Oct	Dynamics: Circular Motion, UCM, Misc. Problems	Ch 6.3	PL #19 Due before class	HW #19 Due at 11:59 pm
20	W	21-Oct	Vector (Cross) Product, Torque	Ch 2.4, 10.6	PL #20 Due before class	HW #20 Due at 11:59 pm
E 2	F	23-Oct	***Exam #2: Lessons 12-19, 1-11***			
21	M	26-Oct	Torque, Rotational Equilibrium, Center of Mass	Ch 10.6, 9.6	PL #21 Due before class	HW #21 Due at 11:59 pm
22	W	28-Oct	Static Equilibrium	Ch 12.1-12.2	PL #22 Due before class	HW #22 Due at 11:59 pm
23	F	30-Oct	Static Equilibrium	Ch 12.1-12.2	PL #23 Due before class	HW #23 Due at 11:59 pm
24	M	2-Nov	Rotations and Angular Kinematics	Ch 10.1-10.3	PL #24 Due before class	HW #24 Due at 11:59 pm
25	W	4-Nov	Angular Kinematics and Rotational Dynamics	Ch 10.4-10.7	PL #25 Due before class	HW #25 Due at 11:59 pm
26	F	6-Nov	Rotational Dynamics	Ch 10.7	PL #26 Due before class	HW #26 Due at 11:59 pm
27	M	9-Nov	Intro to Work and Energy, Dot Product	Ch 2.4, 7.1	PL #27 Due before class	HW #27 Due at 11:59 pm
	W	11-Nov	**No Class - Veterans Day**			

Engineering Physics I - Fall 2026 - Dr. Colón

Lec	Day	Date	Material Covered	Reading Assignments	Pre-Lecture Due	Homework Due
E 3	F	13-Nov	***Exam #3: Lessons 20-26, 1-19***			
28	M	16-Nov	Work-Kinetic Energy Theorem	Ch 7.2-7.3	PL #28 Due before class	HW #28 Due at 11:59 pm
29	W	18-Nov	Potential Energy, Gravitational Potential Energy	Ch 8.1-8.2	PL #29 Due before class	HW #29 Due at 11:59 pm
30	F	20-Nov	Conservation of Energy	Ch 8.3-8.5	PL #30 Due before class	HW #30 Due at 11:59 pm
31	M	23-Nov	Impulse and Momentum	Ch 9.1-9.2	PL #31 Due before class	HW #31 Due at 11:59 pm
	W	25-Nov	**No Class - Thanksgiving Break**			
	F	27-Nov	**No Class - Thanksgiving Break**			
32	M	30-Nov	Conservation of Momentum, Collisions	Ch 9.3-9.5	PL #32 Due before class	HW #32 Due at 11:59 pm
33	W	2-Dec	Misc. Problems with Energy, Momentum	Ch 9.2-9.4	PL #33 Due before class	HW #33 Due at 11:59 pm
34	F	4-Dec	Power; Angular Momentum	Ch 7.4, 11.2-11.3	PL #34 Due before class	HW #34 Due at 11:59 pm
35	M	7-Dec	From Forces to Energy; Last iclicker Problem		PL #35 Due before class	HW #35 Due at 11:59 pm
E 4	W	9-Dec	***Exam #4, Lessons 27-35***			HW #36 Due at 11:59 pm
36	F	11-Dec	Review			

Remarks:

- Assignments (both Pre-Lecture and Homework) are numbered by the lecture (lesson) day.
- Pre-Lecture assignments include material relevant to their *assigned* lecture *day*. For example, PL #05 comprises questions from material in Ch. 5.6-5.7, which is planned for day #05.
- Homework assignments generally include material covered up to the *previous lesson* and none from the due date. For example, HW #05 includes material covered up to lecture day #04.